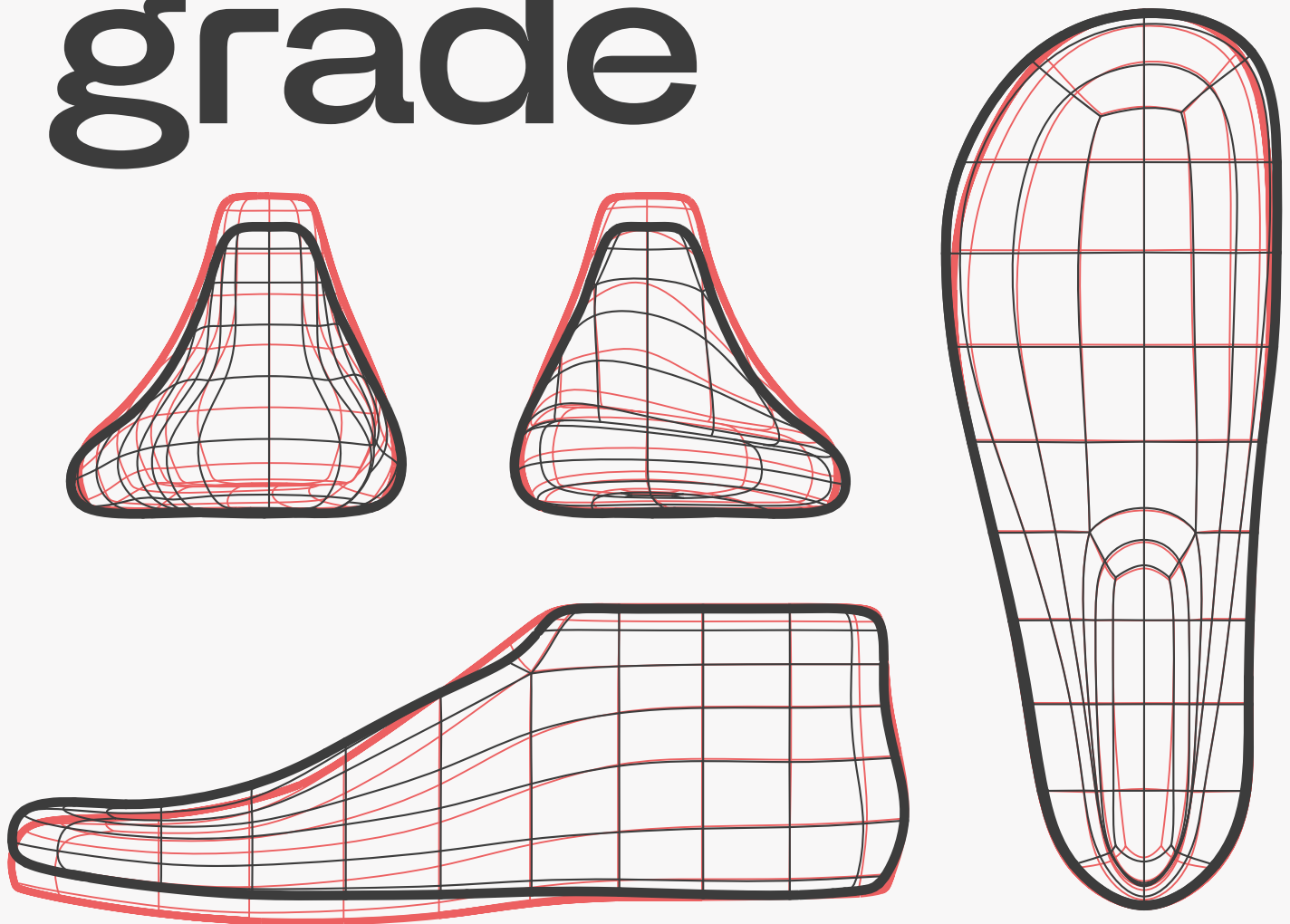


new world grade



Open-Source Global
Grading & Last Library

This work is dedicated to designers, manufacturers and independent developers worldwide who seek to establish their practice upon a unified baseline rooted in international standards rather than arbitrary market conventions. The footwear industry has long been constrained by gatekeeping mechanisms that prioritize market pull over human biomechanical evolution, resulting in persistent inconsistencies in grading and sizing—even within individual brand portfolios. Our objective is to democratize access to standardized information that has historically remained dispersed or proprietary, providing a foundation for improved foot health outcomes, evidence-based design decisions, and collaborative advancement across the sector. By anchoring this resource in international standards and contemporary scientific research on foot biomechanics, we deliberately circumvent the derivative compromises imposed by legacy manufacturing processes and outdated business paradigms that continue to define conventional last morphology.

*Cheers,
The Future Footwear Department*

INTRODUCTION

The foundation of any footwear product rests upon the last—a three-dimensional representation of the human foot that serves as the template for all subsequent construction processes. Despite its fundamental importance, the last design process has remained remarkably resistant to standardization, with practitioners often relying on proprietary systems, empirical traditions, or commercially motivated specifications that lack rigorous biomechanical validation. This document addresses this deficiency by providing comprehensive, standards-based last specifications accompanied by production-ready digital assets compatible with contemporary footwear modeling platforms, including both subdivision surface and NURBS methodologies. While the Mondopoint system represents a significant advancement in international foot measurement standardization, its scope remains limited to length-based classification, offering no systematic guidance for critical dimensional parameters such as ball width or instep height. Our reference system extends beyond these limitations, integrating standardized measurements for last width and instep height alongside longitudinal dimensions. These specifications are derived from international anthropometric standards and validated biomechanical research, ensuring that dimensional relationships reflect actual human foot morphology rather than manufacturing expediency or market-driven aesthetic preferences. The resulting framework provides practitioners with scientifically grounded tools for creating footwear that respects both functional requirements and morphological diversity.

WHAT'S THIS?

This document presents a comprehensive last reference system built upon international anthropometric standards and contemporary biomechanical research.

Production-Ready Digital Geometry

The core offering comprises production-ready digital last models optimized for immediate integration into professional footwear design workflows, delivered in subdivision surface. Each last geometry has been constructed to meet professional-grade specifications: manifold topology free from non-manifold edges, watertight construction suitable for manufacturing processes, and mesh integrity verified at every vertex connection. The models incorporate systematic edge flow architecture that follows the anatomical contours of the human foot, maintaining uniform spacing and predictable subdivision behavior across multiple refinement levels. This careful topological organization ensures that the lasts perform reliably in both digital visualization and physical production contexts, scaling seamlessly from high-resolution rendering to CNC machining operations.

Anthropometric Measurement Framework

Beyond the digital assets themselves, this publication provides detailed dimensional charts that extend the Mondopoint classification system by incorporating ball width and instep height measurements—parameters essential for proper fit but traditionally absent from length-only sizing schemes. These measurements are organized according to international standards, specifically referencing ISO 9407:2019 for foot dimensions and related classification systems. The charts establish clear relationships between foot length, ball girth, and instep height across the complete size range, enabling designers to select appropriate last dimensions based on target demographic characteristics rather than arbitrary brand-specific conventions.

System Applications

Technical specifications for each last include vertex count, polygon distribution, edge flow patterns, and subdivision level recommendations, ensuring that practitioners can make informed decisions regarding model selection for specific applications. The system is designed to function as both a practical tool for immediate product development and an educational resource for understanding the biomechanical principles underlying proper last design, thereby serving both immediate commercial needs and long-term professional development objectives.

Grading Architecture and Mondopoint Progression

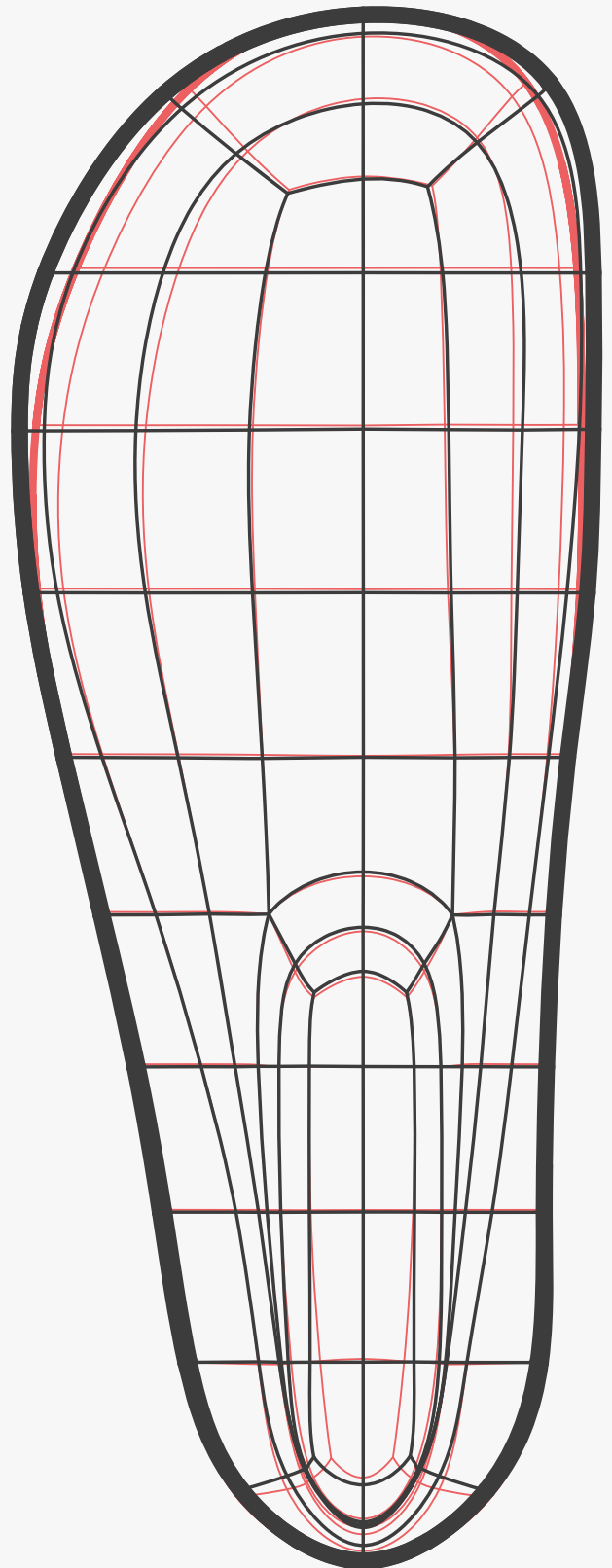
The New World Grading system implements a standardized 10mm length increment per size, following contemporary Mondopoint progression principles that align with World Grading standards. This represents a departure from traditional fractional sizing schemes (6.67mm European, 8.47mm US) in favor of a decimal-based system that simplifies international commerce and manufacturing standardization. At each 10mm length increment, last width increases 2.635 mm, while instep height progresses 1.3mm (or 0.4mm if the footlength is more than 255mm) per half size. These dimensional relationships are derived from anthropometric regression analysis demonstrating that foot dimensions do not scale uniformly: vertical dimensions (height) scale more conservatively than horizontal dimensions (length, width), necessitating non-proportional grading algorithms. Critical measurements that drive grading transformations include maximum instep height at position 55% of last length, and heel width at posterior tangent point. The 10mm progression system facilitates straightforward metric calculation, simplifies inventory management, and provides more intuitive sizing communication to end consumers while maintaining anthropometrically valid dimensional relationships across the complete size range.

Range Definition and Expansion Roadmap

This initial publication phase addresses healthy adult feet within the core anthropometric range, specifically covering foot lengths from 220mm to 320mm in 5mm increments (half sizes). This range encompasses approximately EU sizes 35-50, US men's sizes 4-16, and US women's sizes 5.5-17.5, representing the majority of the adult population across global markets. The phased publication approach prioritizes this core range to establish the foundational system and enable validation through the broadest possible user base.

Subsequent publication phases will expand coverage systematically across multiple dimensions: extended adult sizes beyond 320mm foot length will be released following validation of the core range grading relationships, ensuring that dimensional progressions remain anthropometrically valid at larger scales. The existing three-tier Fit Type classification (Wide, Casual, Bunionizer) provides primary volumetric categorization addressing the majority of fitting requirements. However, for applications requiring finer transverse dimensional control, a secondary width grading layer using Barleycorn letter designations may be implemented within each Fit Type. This would create a matrix of options—for instance, Casual fit available in B through D widths, or Wide fit extending from C through E widths—enabling precise specification while maintaining the conceptual clarity of the primary taxonomy. Implementation of this granular width system will be driven by validated user demand and anthropometric distribution analysis.

Pediatric sizing (foot lengths below 220mm) will constitute a separate publication phase, as children's foot morphology exhibits distinct proportional relationships and growth patterns that require specialized anthropometric consideration and modified grading algorithms. This sequential approach ensures that each size range receives appropriate attention to developmental and morphological characteristics rather than applying simple extrapolation from adult dimensions.



LAST TAXONOMY

This classification system organizes our last collection across **three primary dimensions**: Fit Type, Collar Height, and Last Type.

Fit Type addresses forefoot and midfoot volume requirements, ranging from **Wide** (generous toe splay and ball width for natural foot function) through **Casual** (balanced proportions for everyday comfort) to **Bunionizer** (narrow traditional fit following conventional industry standards).

Collar Height defines ankle coverage with Low-top, Mid-top, and High-top variants, each optimized for specific footwear applications and aesthetic preferences.

Last Type distinguishes biomechanical philosophy: Minimal designs feature zero drop and zero insole allowance for natural foot function, while Sneaker variants incorporate traditional

8mm heel elevation and comfort-oriented geometry. The **highlighted cells** indicate currently published models—Low-Top 00 (Wide Minimal Low-top) and Mid-Top 80 (Casual Sneaker Mid-top)—with additional configurations planned for subsequent release phases.

This matrix enables designers to select the **optimal last** configuration for their **specific application requirements**, combining functional characteristics with target demographic and use-case considerations.

Future releases may introduce Barleycorn letter width variations (A-E) within each Fit Type category to provide additional transverse dimensional options as demand requires.

Fit Type	Collar Height	Last Type	
		Minimal	Sneaker
WIDE	Low	Wide Minimal Low-top	Wide Sneaker Low-top
	Mid	Wide Minimal Mid-top	Wide Sneaker Mid-top
	High	Wide Minimal Hi-top	Wide Sneaker Hi-top
CASUAL	Low	Casual Minimal Low-top	Casual Sneaker Low-top
	Mid	Casual Minimal Mid-top	Casual Sneaker Mid-top
	High	Casual Minimal Hi-top	Casual Sneaker Hi-top
BUNIONIZER	Low	Bunionizer Minimal Low-top	Bunionizer Sneaker Low-top
	Mid	Bunionizer Minimal Mid-top	Bunionizer Sneaker Mid-top
	High	Bunionizer Minimal Hi-top	Bunionizer Sneaker Hi-top

MORE INCOMING

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APPENDIX

This appendix provides comprehensive technical resources for implementing the New World Grading system, including digital asset specifications, dimensional charts, and access protocols for the complete last library. Each asset entry includes detailed datasheets specifying topology characteristics, file formats, and recommended applications. Direct hyperlinks to the official GitHub repository provide immediate access to production-ready files and documentation.

The digital assets and grading specifications represent a living system subject to continuous refinement based on emerging research, manufacturing feedback, and validation data. Users should reference the GitHub repository for current versions, as ongoing development may introduce specification improvements. The dimensional specifications and last geometries are derived from extensive analysis of international anthropometric databases, biomechanical literature, and iterative design development conducted over multiple years. Formal validation through clinical assessment and population studies remains ongoing and will be documented in subsequent publications. This validation incorporates community feedback as a critical component, recognizing that real-world application across diverse populations provides insights unattainable through laboratory protocols alone. We encourage developers, manufacturers, and users to contribute observations through the project repository, fostering collaborative development. This evidence-driven approach acknowledges that optimal specifications emerge through iterative validation as population characteristics and biomechanical knowledge evolve.

LOW-TOP 00

Asset Datasheet

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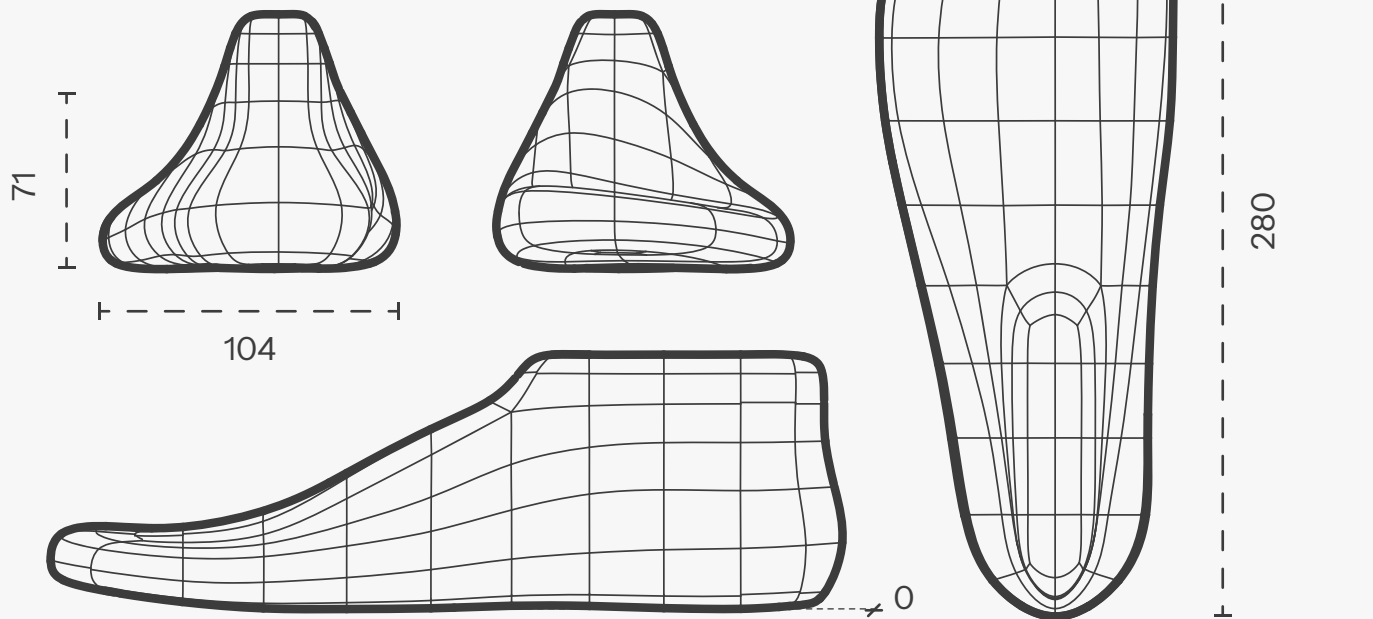
Barefoot style last with zero drop and zero insole allowance, making it an ideal candidate for flexible 3d printed shoes.

ASSET

Name: **Low-Top 00**

Size: **280-104-71**

File: **The Low-Top 80 (280-104-71).obj**



FIT AND USE

The shape is thought for recovery and activity at the same time. Is more supportive in the rear foot to allow for some athletic movements while allowing natural toe splay and the use of toe spacers. Minimal toe spring and anatomically sculpted bottom for natural movement in the foot-ground contact.

Best for shoes without socks

Forefoot: wide fit with a generous toe splay

Midfoot: snug fit for sock-less wear with contoured arch

Heel: zero stack height and snug fit for responsive movements

Ankle: low-top height with a tighter fit to avoid ankle slip



GET IT HERE

access to the 3d model is granted free of charge, under MIT license, allowing designers, artisans and manufacturers to integrate it into their design and production workflows.

CHART DROP

Mondo-point	Last Dimensions			Conversion	
	Length	Width	Instep Height	EU	US (M)
220	235	92.0	60.5	35	4
225	240	93.3	61.8	36	4.5
230	245	94.7	63.1	36.5	5
235	250	96.1	64.4	37	5.5
240	255	97.5	65.7	38	6.5
245	260	98.8	67.0	38.5	7
250	265	100.2	68.3	39.5	7.5
255	270	101.6	69.6	40	8
260	275	103.0	70.9	41	8.5
265	280	104.3	71.3	41.5	9
270	285	105.7	71.7	42.5	9.5
275	290	107.1	72.1	43	10
280	295	108.5	72.5	44	10.5
285	300	109.8	72.9	44.5	11
290	305	111.2	73.3	45.5	11.5
295	310	112.6	73.7	46	12
300	315	114.0	74.1	47	13
305	320	115.3	74.5	47.5	13.5
310	325	116.7	74.9	48	14
315	330	118.1	75.3	48.5	14.5
320	335	119.5	75.7	49	15

The chart above provides complete dimensional specifications for the **Low-Top 00** last from foot length **220mm to 320mm**. Each size includes three critical measurements: last length , last width at ball girth position, and **instep height at 55%** foot length, all specified in **millimeters**. For convenience, equivalent EU and US men's sizes are provided in the conversion columns. Dimensions are based on international anthropometric standards (ISO 9407:2019) and represent the actual last measurements, not foot measurements. The highlighted reference size (265mm) serves as the baseline for grading validation.

MID-TOP 80

Asset Datasheet

Sneaker style last with 8 mm drop and zero insole allowance, making it an ideal candidate for commercial 3d printed shoes.

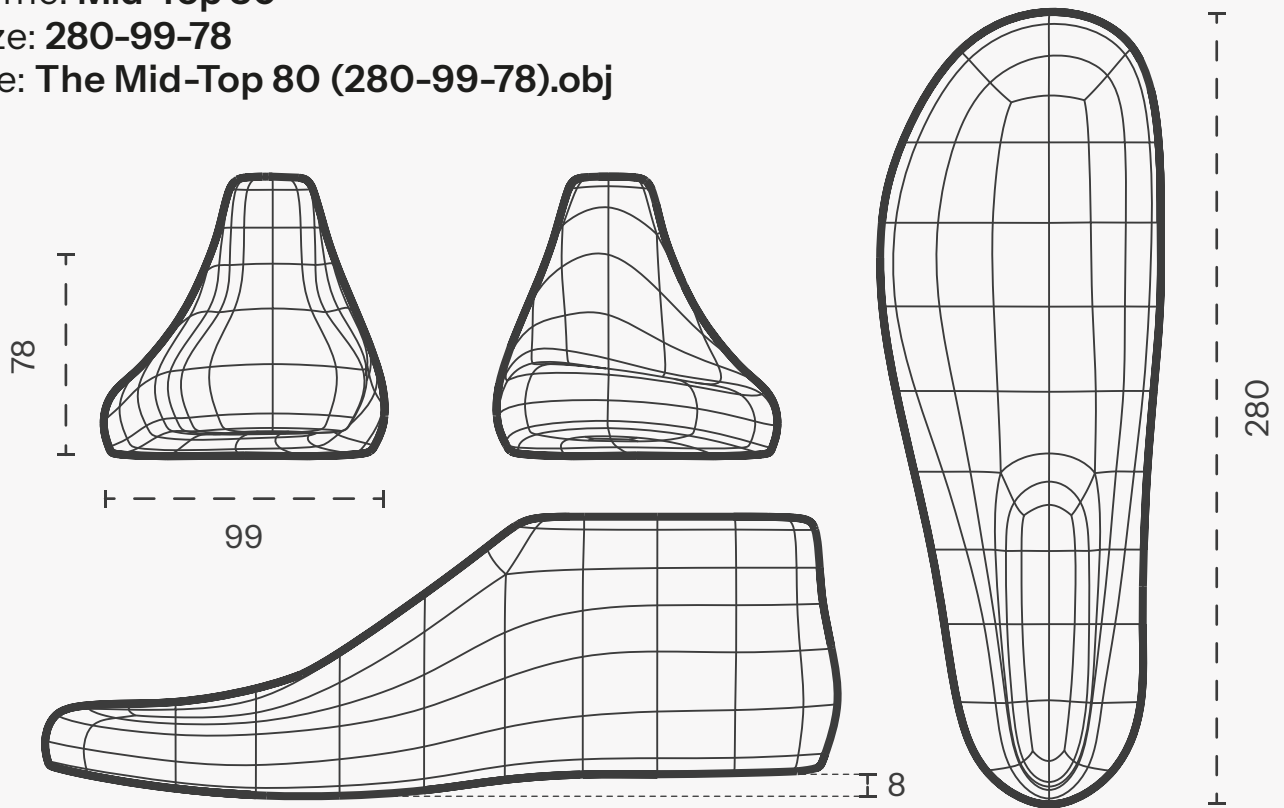
ASSET

Name: **Mid-Top 80**

Size: **280-99-78**

File: **The Mid-Top 80 (280-99-78).obj**

[mm]



FIT AND USE

The shape is designed for everyday comfort with a familiar sneaker feel. Features an 8mm heel drop that provides traditional sneaker comfort and a casual walking experience. The zero insole allowance creates a precise, direct fit optimized for 3d printed materials. Offers classic sneaker aesthetics and comfort with improved fit considerations for better daily wear and enhanced comfort.

Best for shoe with socks.

Forefoot: standard fit with a familiar toe splay

Midfoot: comfort fit to wear with socks

Heel: traditional stack height, with familiar fit and cushioning

Ankle: mid-top height for style and stability



GET IT HERE

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CHART DROP

Mondo-point	Last Dimensions			Conversion	
	Length	Width	Instep Height	EU	US (M)
220	235	86.9	67.3	35	4
225	240	88.2	68.6	36	4.5
230	245	89.6	69.9	36.5	5
235	250	91.0	71.2	37	5.5
240	255	92.4	72.5	38	6.5
245	260	93.7	73.8	38.5	7
250	265	95.1	75.1	39.5	7.5
255	270	96.5	76.4	40	8
260	275	97.9	77.7	41	8.5
265	280	99.2	78.1	41.5	9
270	285	100.6	78.5	42.5	9.5
275	290	102.0	78.9	43	10
280	295	103.4	79.3	44	10.5
285	300	104.7	79.7	44.5	11
290	305	106.1	80.1	45.5	11.5
295	310	107.5	80.5	46	12
300	315	108.9	80.9	47	13
305	320	110.2	81.3	47.5	13.5
310	325	111.6	81.7	48	14
315	330	113.0	82.1	48.5	14.5
320	335	114.4	82.5	49	15

The chart above provides complete dimensional specifications for the **Mid-Top 80** last from foot length **220mm to 320mm**. Each size includes three critical measurements: last length , last width at ball girth position, and **instep height at 55%** foot length, all specified in **millimeters**. For convenience, equivalent EU and US men's sizes are provided in the conversion columns. Dimensions are based on international anthropometric standards (ISO 9407:2019) and represent the actual last measurements, not foot measurements. The highlighted reference size (265mm) serves as the baseline for grading validation.

links

FULL REPOSITORY

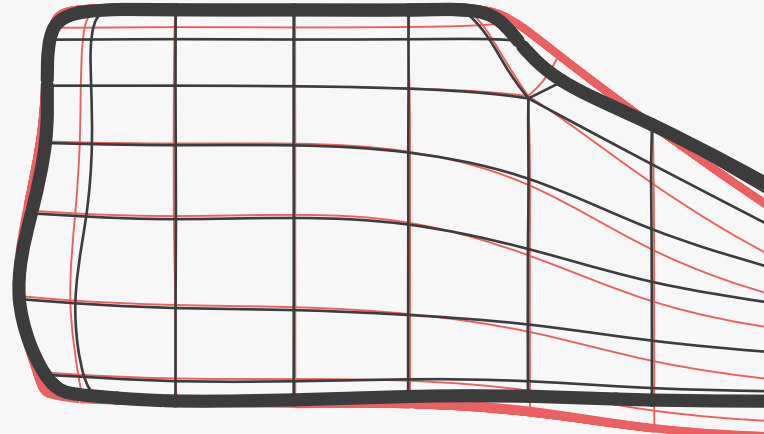
<https://github.com/AffinitySpace/new-world-grade>

MINIMAL

<https://github.com/AffinitySpace/new-world-grade/tree/minimal>

SNEAKER

<https://github.com/AffinitySpace/new-world-grade/tree/sneaker>

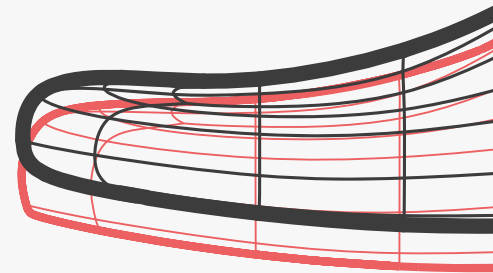


credits

Fabrizio De Lucia, Jared Max Fiorovich
& The Future Footwear Department

Special thanks to:

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